

Scalability & Future Plan

Date	02 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Current System Limitations :

S.N o	Limitation	Impact	Priority to Address
1	Dataset is processed locally using a Jupyter Notebook and is not optimized for very large-scale datasets.	Increased execution time and higher memory usage when processing millions of records.	High
2	The model is currently a standalone implementation without a web interface or REST API.	End users cannot access the prediction system through a browser or external applications.	Medium
3	The dataset is highly imbalanced, resulting in lower recall for the minority class (bad customers).	Some risky applicants may not be identified accurately.	High

Scalability Plan :

S.N o	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports single-user execution on a local machine.	Deploy the model as a Flask/FastAPI service with cloud hosting to support multiple concurrent users.
2	Data Storage	Uses CSV files for storing application and credit	Integrate a relational database such as MySQL or PostgreSQL for

		records.	efficient data management and scalability.
3	Performance	Model executes locally with acceptable performance for medium-sized datasets.	Optimize preprocessing, implement batch prediction, and use cloud infrastructure for faster execution.
4	Security	No authentication or secure data transmission implemented.	Add user authentication, HTTPS communication, role-based access control, and data encryption for secure deployment.

Future Roadmap :

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the trained model using Flask or Streamlit with a web-based user interface.	1–2 Months	Enables real-time credit approval prediction for users through a web application.
Phase 2	Integrate a SQL database and automated data ingestion pipeline.	2–3 Months	Improves scalability, reliability, and efficient storage of customer records.
Phase 3	Improve model performance using SMOTE, XGBoost, LightGBM, and automated hyperparameter optimization.	3–4 Months	Enhances prediction accuracy, recall, and overall model robustness.
Phase 4	Deploy the application on cloud platforms (AWS, Azure, or Google Cloud) with REST APIs and monitoring dashboards.	4–6 Months	Provides enterprise-level scalability, remote accessibility, and continuous monitoring.